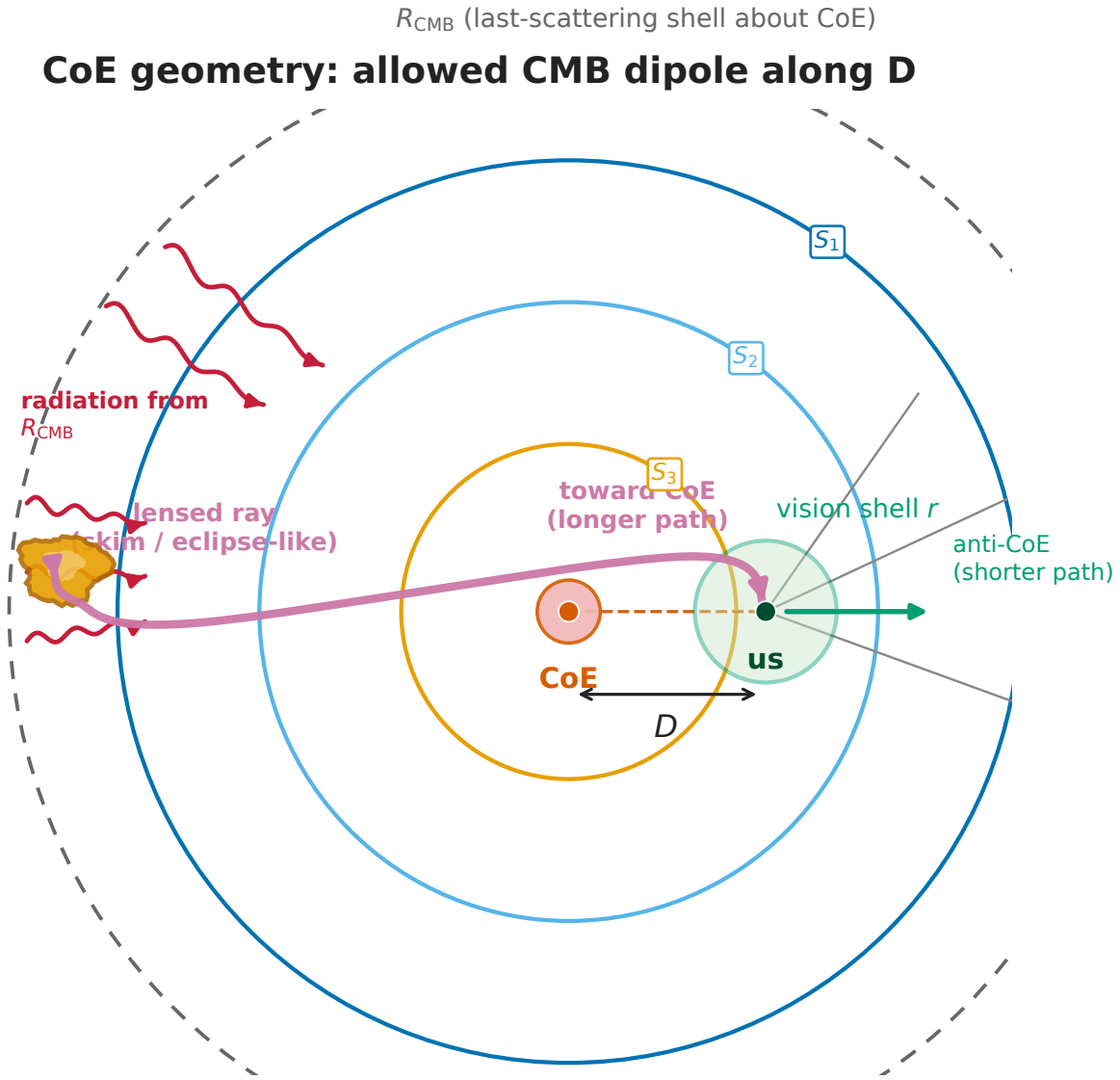

Off-center CoE geometry (schematic). CMB from $R_{CMB} \gg D$; vision shell radius r; $S_n \to S_{n+1}$ spacing $2r$; offset D. Opposite sides of **D** ⇒ geometric CMB dipole; lensing (orange) near the CoE-facing sky warps paths (purple skim / eclipse-like ray). Late Hubble quadrupole lives on cluster /r scales (not shown as CMB quad).